IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

**Corrected Copy**

**MATHEMATICS 2B (E-STREAM AND I-STREAM)**

Friday, 24 May 2:00 pm

Time allowed: 1:30 hours

**There are TWO questions on this paper.**

**Answer TWO questions.**

*Question 1 carried 60% of the marks. Questions 2 carries 40% of the marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : B. Clerckx
Second Marker(s) : B.K.W. Ee

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## THE QUESTIONS

[25]

1. a) Consider two discrete random variables $X$ and $Y$ that have the joint probability mass function represented by the joint probability table

| | | $Y$ | | |
|---|---|---|---|---|
| | | 0 | 1 | 2 |
| $X$ | 0 | 0.2 | 0.1 | 0.1 |
| | 1 | 0.0 | 0.1 | 0.2 |
| | 2 | 0.1 | 0.1 | 0.1 |

i) Compute the probability that $X$ is smaller or equal to $Y$, i.e. $P(X \leq Y)$ and the probability that $X$ is strictly smaller than $Y$, i.e. $P(X < Y)$.

[2]

ii) Compute the marginal probability mass function of $X$ and $Y$.

[2]

iii) Compute the expectation of $X$, i.e. $E(X)$, and the expectation of $Y$, i.e. $E(Y)$.

[2]

iv) Compute the variance of $X$ and the variance of $Y$, i.e. $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$, the covariance between $X$ and $Y$, i.e. $\mathrm{Cov}(X,Y)$, and the correlation coefficient between $X$ and $Y$, i.e. $\mathrm{Corr}(X,Y)$.

[4]

v) Are $X$ and $Y$ uncorrelated? Independent? Provide your reasoning.

[2]

vi) Compute the conditional probability mass function of X given that $Y = 0,1,2$.

[3]

vii) Compute the conditional expectation of X given that $Y = 0,1,2$.

[3]

viii) By making use of vii), compute the expectation of $X$, i.e. $E(X)$.

[2]

b) Assume $X \sim N(1,4)$ and take $Y = e^X$.

i) Compute the probability $P(Y \leq e)$.

[2]

ii) Compute the probability density function of Y, i.e. $f_Y(y)$.

[3]

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[25]

2. a) Consider two exponentially distributed random variables with parameter $\lambda > 0$, i.e. $X \sim EXPO(\lambda)$ and $Y \sim EXPO(\lambda)$. $X$ and $Y$ are independent.

i) Compute the joint probability density function $f_{X,Y}(x,y)$.

[4]

ii) Make the change of variables $U = \frac{X}{X+Y}$ and $V = X$ and compute the joint probability density function $f_{U,V}(u,v)$.

[4]

iii) Compute the marginal probability density function of $U$ and $V$, i.e. $f_U(u)$ and $f_V(v)$.

[4]

iv) Are U and V independent? Provide your reasoning.

[3]

b) Consider two continuous random variables $X$ and $Y$ with joint probability density function $f_{X,Y}(x,y)$.

i) Prove that $E(E(Y|X)) = E(Y)$. Provide your reasoning. [4]

ii) Prove that

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (E(X|Y=y) - E(X))(x - E(X|Y=y)) f_{X,Y}(x,y)\, dx\, dy = 0.$$

Provide your reasoning. [6]

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